

Research Summary

Title : Robust Aggregation Against Byzantine Attacks in Federated Learning for Internet of Medical Things (IoMT)-Based Smart Healthcare.

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1. Problem Definition

Federated Learning (FL) has become a promising artificial intelligence paradigm for smart healthcare because it enables multiple hospitals, clinics, and Internet of Medical Things (IoMT) devices to collaboratively train machine learning models without sharing raw patient data. This decentralized learning process improves privacy and helps organizations comply with regulations such as GDPR and HIPAA. However, FL remains highly vulnerable to Byzantine attacks, where compromised or malicious participants intentionally upload manipulated model updates during the training process. These poisoned updates can significantly reduce model accuracy, corrupt the global model, and even cause complete training failure.

In IoMT-enabled healthcare systems, Byzantine attacks are particularly dangerous because medical devices such as wearable ECG monitors, glucose sensors, and remote patient monitoring systems continuously contribute data for model training. If attackers manipulate these updates, AI models may generate incorrect predictions, delayed diagnoses, or inappropriate treatment recommendations. Existing aggregation algorithms such as FedAvg are not designed to detect malicious updates and therefore cannot guarantee reliable learning in adversarial healthcare environments.

Clinical Impact: Solving the Byzantine attack problem is critical for ensuring trustworthy AI-assisted healthcare. A secure federated learning framework can improve diagnostic reliability, support accurate disease prediction, and enhance remote patient monitoring while preserving patient privacy. Robust aggregation methods also reduce the risk of false clinical decisions, improve collaboration among healthcare institutions, and increase confidence in AI-based medical applications. As IoMT devices continue to expand in hospitals and home healthcare, developing Byzantine-resilient federated learning systems has become essential for protecting patient safety and ensuring dependable clinical outcomes.

2. State-of-the-Art (SOTA) Analysis

Researchers have proposed several robust aggregation techniques to defend Federated Learning (FL) against Byzantine attacks in healthcare environments. These methods aim to identify and remove malicious model updates before generating the global model while maintaining high prediction accuracy and preserving patient privacy. Krum is one of the earliest robust aggregation

algorithms that selects the model update closest to the majority of participants. It is effective against a limited number of Byzantine clients but its performance decreases when the number of malicious participants increases. Trimmed Mean improves robustness by removing the highest and lowest parameter values before aggregation.

It performs better than traditional FedAvg under moderate attack scenarios and has relatively low computational complexity. However, its effectiveness declines when attackers coordinate sophisticated poisoning strategies. Bulyan combines multiple robust aggregation techniques to achieve stronger protection against adversarial updates. It provides high model reliability and excellent attack resistance but introduces significant computational overhead, making it less suitable for resource-constrained IoMT devices. Recent studies have also combined Differential Privacy (DP) with robust aggregation.

This approach simultaneously protects patient privacy and defends against malicious clients. However, the injected privacy noise may reduce model accuracy, especially for complex medical imaging and diagnostic tasks. More recent research focuses on adaptive robust aggregation using anomaly detection and trust-based client evaluation. These methods dynamically identify suspicious participants and improve resilience against evolving Byzantine attacks while maintaining better model performance in heterogeneous healthcare environments.

2.1 Comparison of Recent SOTA Methods (2023–2026)

Method	Year	Accuracy	Privacy	Latency	Main Limitation
Krum	2023	High	Medium	Medium	Performance decreases with many malicious clients
Trimmed Mean	2023	High	Medium	Low	Weak against coordinated poisoning attacks
Bulyan	2024	Very High	Medium	High	High computational complexity
Differential Privacy + FL	2024	Medium–High	Very High	Medium	Privacy noise reduces model accuracy

Method	Year	Accuracy	Privacy	Latency	Main Limitation
Adaptive Robust Aggregation	2025	Very High	High	Medium	Requires efficient anomaly detection and additional computation

Overall, recent studies demonstrate that robust aggregation substantially improves the security of Federated Learning in healthcare. Nevertheless, no single method simultaneously provides maximum robustness, high diagnostic accuracy, low communication overhead, scalability, and strong privacy preservation. These trade-offs indicate that additional research is still required to develop practical Byzantine-resilient Federated Learning frameworks for real-world IoMT-based smart healthcare systems.

3. Technical Gaps

Although significant progress has been made in defending Federated Learning against Byzantine attacks, several important challenges remain unresolved. First, many existing robust aggregation algorithms assume a static network where the number of participating clients remains unchanged throughout the training process. In real IoMT-based healthcare environments, wearable devices, mobile sensors, and hospital systems frequently join and leave the network, making this assumption unrealistic. Second, most current methods focus primarily on security while overlooking the limited computational resources of IoMT devices. Advanced aggregation algorithms such as Bulyan and other trust-based methods require high computational power and memory, which increases energy consumption and training time on resource-constrained medical devices. Third, communication overhead remains a major limitation. Robust aggregation often requires multiple rounds of parameter verification and client evaluation, increasing communication latency between healthcare institutions and the central server. This delay can negatively affect real-time clinical applications such as remote patient monitoring and emergency diagnosis.

Another important challenge is the trade-off between privacy and model accuracy. Integrating Differential Privacy or encryption techniques strengthens data protection but introduces additional noise into model updates, which may reduce diagnostic performance for complex healthcare datasets. Finally, existing methods are often evaluated under limited attack scenarios. Modern

attackers can launch adaptive and coordinated Byzantine attacks that continuously modify their behavior to avoid detection. Current aggregation strategies are not sufficiently robust against these evolving threats, particularly in heterogeneous healthcare environments where data distributions differ significantly across hospitals and IoMT devices.

4. Proposed Direction

A promising solution is the development of a Hybrid Adaptive Byzantine-Resilient Federated Learning Framework for IoMT-based smart healthcare. The proposed framework combines adaptive robust aggregation, lightweight anomaly detection, trust-based client scoring, and Differential Privacy within a single learning architecture. Instead of treating every client equally, the framework continuously evaluates the reliability of each participant based on its historical behavior and update consistency. Suspicious model updates are filtered through lightweight anomaly detection before the aggregation process begins. Trust scores are then used to assign different aggregation weights, allowing reliable healthcare institutions to contribute more significantly to the global model. To reduce communication overhead, edge computing can perform preliminary aggregation near IoMT devices before transmitting compressed model updates to the central server. Differential Privacy is applied adaptively according to the sensitivity of the medical data, minimizing the loss of diagnostic accuracy while preserving patient confidentiality. This hybrid framework is expected to improve Byzantine attack resilience, reduce communication latency, preserve patient privacy, and maintain high diagnostic accuracy, making Federated Learning more practical for large-scale smart healthcare systems.

5. Bibliography

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